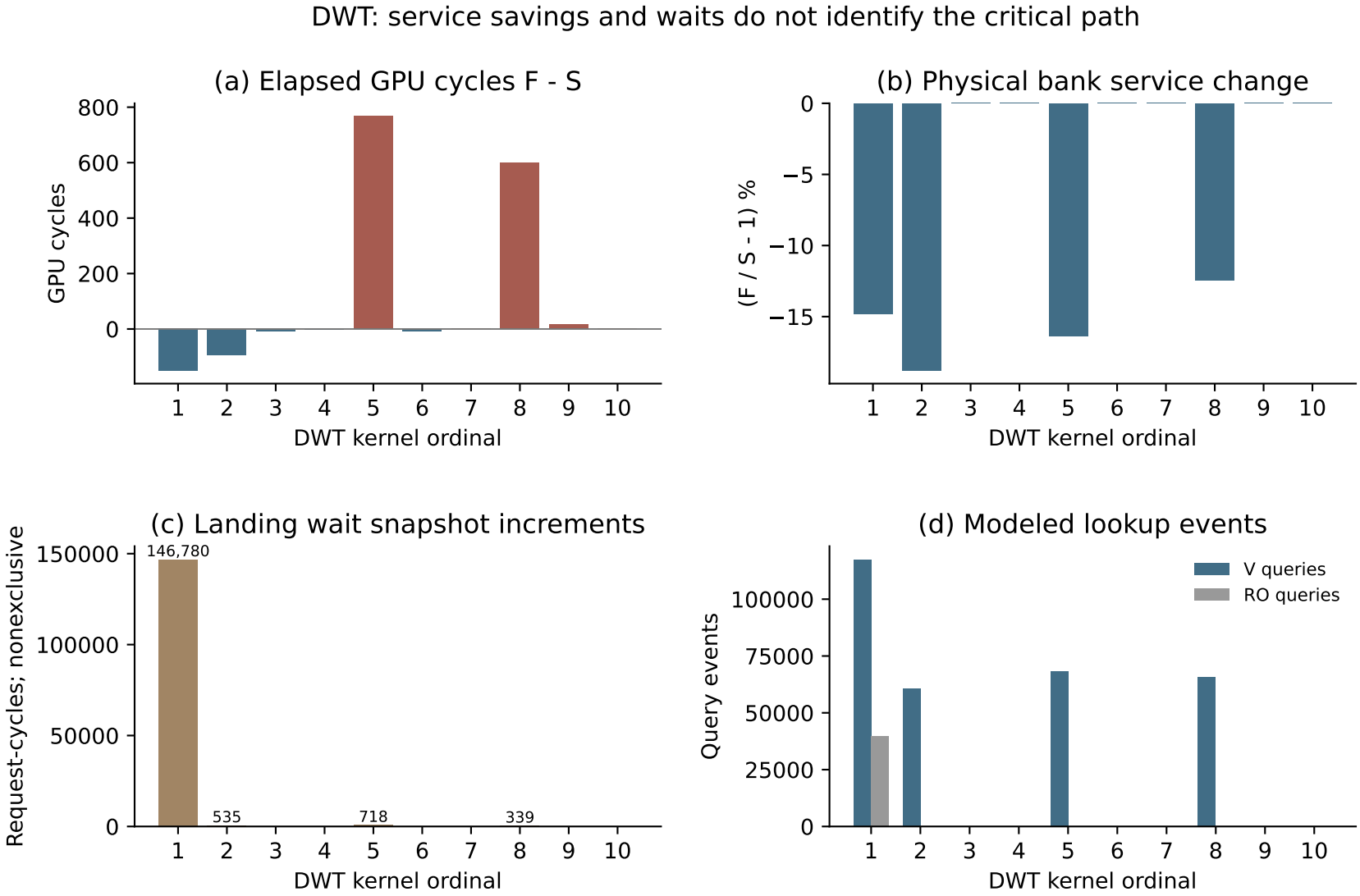
DWT: service savings and waits do not identify the critical path
(a) Elapsed GPU cycles F - S
GPU cycles
800
600
400
200
0
1 2 3 4 5 6 7 8 9 10
DWT kernel ordinal
(b) Physical bank service change
(F / S - 1) %
0
−5
−10
−15
1 2 3 4 5 6 7 8 9 10
DWT kernel ordinal
(c) Landing wait snapshot increments
Request-cycles; nonexclusive
150000
100000
50000
0
146,780
535
718
339
1 2 3 4 5 6 7 8 9 10
DWT kernel ordinal
(d) Modeled lookup events
Query events
100000
75000
50000
25000
0
V queries
RO queries
1 2 3 4 5 6 7 8 9 10
DWT kernel ordinal
98.9% of landing wait increments occur in faster kernel 1.
Kernels 5 and 8 regress with zero new RO queries; indirect prior-state effects remain possible.